

Sample Size Calculations - Cochran-Armitage Test for Trend

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Webpage for the exercises:

http://csg.sph.umich.edu/abecasis/cats/gas_power_calculator/index.html

<http://zzz.bwh.harvard.edu/gpc/cc2.html>

Question 1

For a complex disease study, you plan to collect 35,000 cases and 70,000 controls and wish to know if this is a sufficient sample size to detect associations with disease susceptibility loci. The disease has a population prevalence of 5%. You wish to estimate the power for a genotypic relative risk of 1.2 and a disease allele frequency of 0.02. What is the power for $\alpha=5 \times 10^{-8}$ under a multiplicative model ($\gamma_2 = \gamma_1^2$) a.) _____ and dominant model ($\gamma_2 = \gamma_1$) b.) _____?

Question 2

For your study, you hypothesize that you will try to replicate associations for 100 variants that are in linkage equilibrium and you want to reject the null hypothesis using a p-value of 0.05. What is the Bonferroni correction you should use a.) _____. Determine what your power would be if you used a Bonferroni correction to control for the Family Wise Error Rate (FWER) for testing 100 variants. Using the parameters provided in question 1 but for a sample size of 20,000 cases and 20,000 controls what is the power under the multiplicative model b.) _____ and under a dominant model c.) _____?

Question 3

You determine that you can ascertain 50,000 cases and 50,000 controls what is the power using the same parameters as described in question 1 for the multiplicative model _____ and dominant model _____?

Question 4

The power of the Cochran-Armitage test for trend is dependent on the underlying genetic model. Using the parameters from question 1 which of the following underlying genetic models: multiplicative ($\gamma_2 = \gamma_1^2$), additive ($\gamma_2 = 2\gamma_1 - 1$), dominant ($\gamma_2 = \gamma_1$) or recessive ($\gamma_1 = 1$) would you predict to be the most powerful a.) _____ and least powerful b.) _____?

Question 5

For study design with equal numbers of cases and controls a genotype relative risk of 1.5 under a recessive model for a disease with a population prevalence of 0.05 and disease allele frequency of 0.1. How many cases a.) _____ and controls b.) _____ should you ascertain for $\alpha=5.0 \times 10^{-8}$ and $1-\beta=0.80$? *Use power2 or Genetic Power Calculator, GAS power cannot calculate for more than 100,000 cases.

Question 6

You are performing a rare variant association study and you assume that that cumulative frequency of the causal variants in your gene region is 0.01 with every variant having an effect size of 1.4. The disease you are studying has a prevalence of 5%. For a study with 0.8 power and an $\alpha=2.5 \times 10^{-6}$ under a dominant model for equal numbers of cases and controls what is the total sample size a.) _____ do you need to ascertain. What is the total sample size b.) _____ you need to ascertain if the cumulative frequency of causal variants is only 0.005?